



MANIPAL INSTITUTE OF TECHNOLOGY
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Mini Project Report
of
Big Data Analytics and Tools Lab [DSE 3242]

**GRAPH-BASED SOCIAL NETWORK
ANALYSIS FOR INFLUENCER
DETECTION**

**SUBMITTED
BY**

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Jan-Apr 2026



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SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Manipal

04/04/2026

CERTIFICATE

This is to certify that the project titled Graph-Based Social Network Analysis for Influencer Detection is a record of the bonafide work done by **Shashank Bhat (230968027)**, **Sohan Sanil (230968074)**, submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in DATA SCIENCE & ENGINEERING of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent Institute of Manipal Academy of Higher Education), during the academic year 2025-2026.

Name and Signature of Examiners:

1.

2.

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CHAPTER 1

INTRODUCTION

Social media platforms such as Twitter, Instagram, and LinkedIn generate massive volumes of data through user interactions like likes, comments, shares, and mentions. These interactions form complex relationships that can be modelled as graphs, where users are represented as nodes and interactions as edges. Analyzing such networks helps in identifying influential users, understanding information flow, and detecting communities. However, traditional data processing systems are not capable of handling such large-scale and complex datasets. This project focuses on applying graph-based social network analysis using Big Data tools to detect influential users and communities.

1.1 Objectives

- To preprocess social network data
- To construct a graph-based representation of user interactions
- To apply graph analytics algorithms for influencer detection
- To identify communities within the network

1.2 Dataset Characteristics

The dataset used for this project is a social network dataset obtained from publicly available sources such as Kaggle and SNAP.

The dataset consists of user interactions represented as:

- Source node (user)
- Target node (user)
- Type of interaction (follow, mention, etc.)

Feature	Type	Description
Source	Categorical	User initiating interaction
Target	Categorical	User receiving interaction
Interaction Type	Categorical	Type of relationship

Table 1.1: Dataset Characteristics

The dataset enables analysis of relationships between users and helps in identifying influence patterns within the network.

CHAPTER 2 METHODOLOGY

The project follows a data analytics pipeline consisting of preprocessing, graph construction, analysis, and visualization.

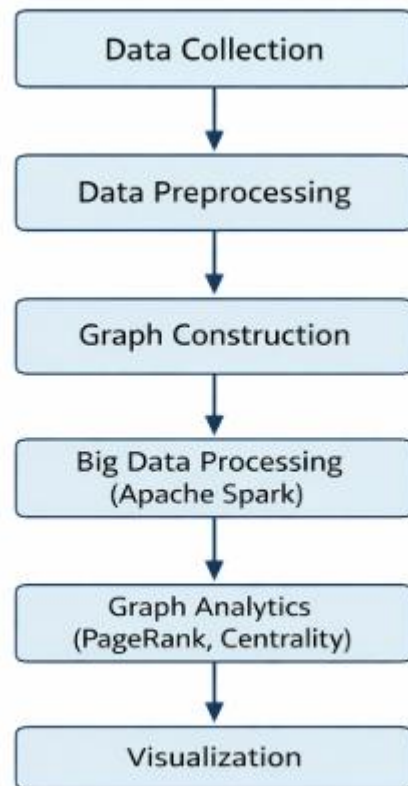


Fig 2.1: Flow of data through various stages of the analytics pipeline

2.1 Preprocessing

Data preprocessing involves:

- Removing duplicate entries
- Handling missing values
- Converting raw data into node-edge format

This ensures the dataset is suitable for graph-based analysis.

2.2 Exploratory Analysis

Basic analysis was performed to understand:

- Distribution of connections
- Network density
- Interaction patterns

This helps in understanding the structure of the dataset before applying graph algorithms.

2.3 Building the Model

The graph was constructed where:

- Nodes represent users
- Edges represent interactions

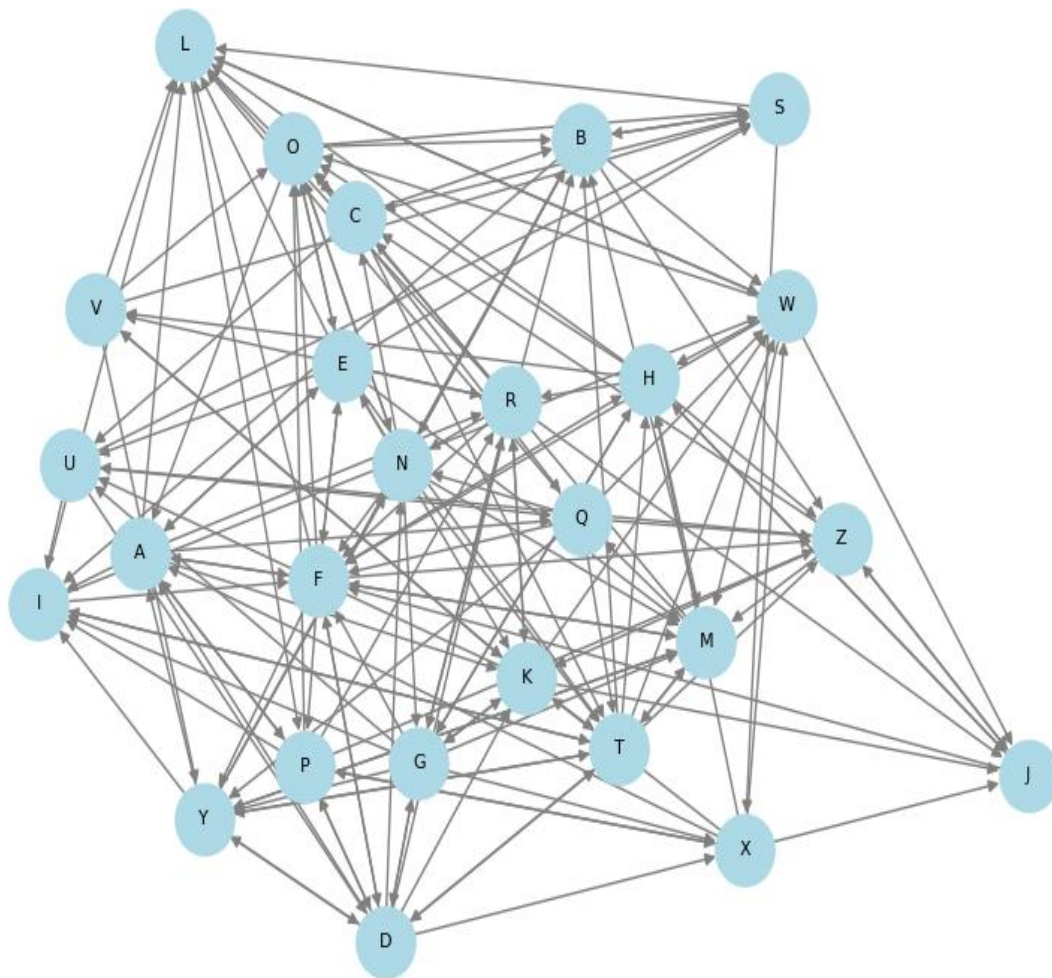


Figure 2.2: Graph Representation of Social Network

Graph analytics algorithms used:

- PageRank
- Degree Centrality
- Betweenness Centrality
- Community Detection

Big Data processing was performed using Apache Spark.

```
shashank@FiRe:~$ hdfs dfsadmin -report
2026-04-17 11:55:51,960 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 994068325472 (925.80 GB)
DFS Remaining: 972425666560 (905.64 GB)
DFS Used: 21642658912 (20.16 GB)
DFS Used%: 2.18%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: FiRe.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 21642658912 (20.16 GB)
Non DFS Used: 32040496032 (29.84 GB)
DFS Remaining: 972425666560 (905.64 GB)
DFS Used%: 2.00%
DFS Remaining%: 89.95%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Apr 17 11:55:50 IST 2026
Last Block Report: Fri Apr 17 11:37:59 IST 2026
```

Figure 2.3: HDFS Storage Report

```
shashank@FiRe:~$ ssh shashank@10.41.72.119
The authenticity of host '10.41.72.119 (10.41.72.119)' can't be established.
ED25519 key fingerprint is SHA256:xbMgLfSnftHlV/bE8507D5QL6WPG+8n2v4I+6WGFq+8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.41.72.119' (ED25519) to the list of known hosts.
shashank@10.41.72.119's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 17 06:06:04 UTC 2026

System load:  0.08                Processes:            29
Usage of /:   0.4% of 1006.85GB   Users logged in:     1
Memory usage: 20%                IPv4 address for eth0: 172.25.205.42
Swap usage:   0%

shashank@DESKTOP-3VG48H1:~$ start-
```

Figure 2.4: Remote Server Access

2.4 Model Evaluation and Comparison

Different graph algorithms were evaluated based on:

- Influence detection capability
- Connectivity analysis
- Community detection accuracy

Each algorithm provides different insights into the network structure.

CHAPTER 3

RESULT ANALYSIS

The results obtained from graph analytics are presented below.

3.1 PageRank Analysis

```
shashank@FiRe:~$ python3 project.py
=== PageRank (Influence Score) ===
F 0.06
W 0.055
L 0.054
B 0.052
T 0.052
O 0.049
M 0.049
Z 0.048
C 0.047
I 0.041
H 0.04
J 0.036
P 0.036
N 0.035
Y 0.035
A 0.035
D 0.034
K 0.033
S 0.033
U 0.032
Q 0.03
R 0.027
E 0.025
G 0.024
X 0.022
V 0.017
```

Figure 3.1: PageRank Results

User F has the highest PageRank score (0.06), indicating it is the most influential user.

3.2 Degree Centrality Analysis

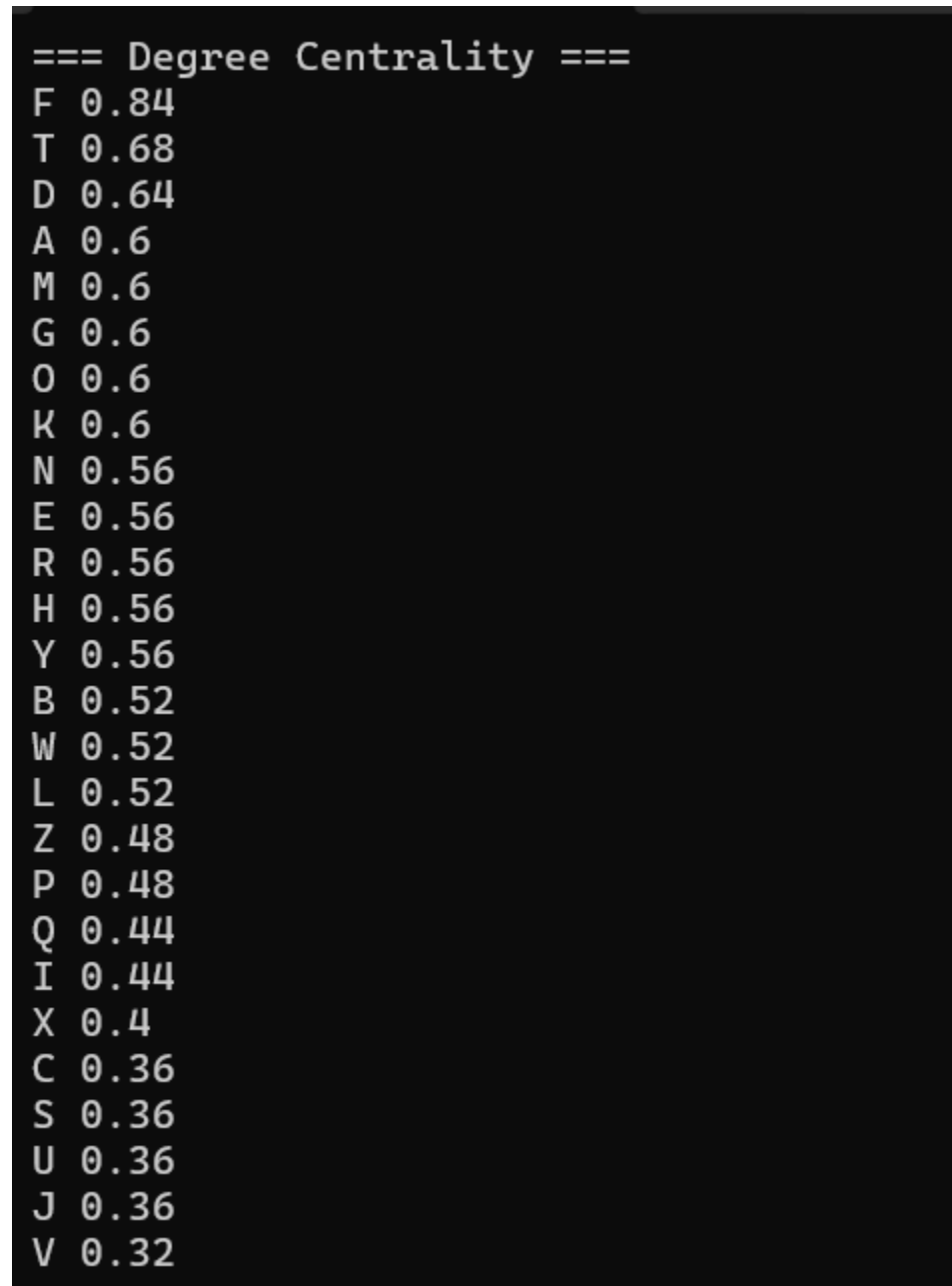


Figure 3.2: Degree Centrality Results

User F has the highest degree centrality (0.84), showing maximum connectivity.

3.3 Community Detection

```
=== Communities ===
Community 1: ['E', 'O', 'N', 'U', 'B', 'S', 'C']
Community 2: ['R', 'V', 'G', 'H', 'J', 'K', 'Z']
Community 3: ['M', 'W', 'Y', 'I', 'Q', 'L', 'T']
Community 4: ['F', 'X', 'D', 'A', 'P']
shashank@FiRe:~$ nano project.py
shashank@FiRe:~$ nano project.py
shashank@FiRe:~$ python3 project.py
=== PageRank (Influence Score) ===
F 0.06
W 0.055
```

Figure 3.3: Community Clusters

The network is divided into multiple communities, showing group structures.

3.4 Betweenness Centrality

```
Top Connectors (Bridges):
F 0.099
O 0.061
T 0.061
D 0.059
H 0.056
shashank@FiRe:~$ |
```

Figure 3.4: Betweenness Centrality

User F acts as a bridge between communities.

3.5 Overall Analysis

The results show that user F consistently ranks highest across multiple metrics, confirming it as the most influential node in the network.

CHAPTER 4

CONCLUSION

This project demonstrates the effectiveness of combining Big Data technologies and graph analytics for analysing social networks.

The system successfully identifies influential users and communities using graph-based techniques. Big Data tools enable efficient handling of large datasets, making the approach scalable.

However, limitations include data availability and processing complexity. Future work can involve real-time data processing and advanced techniques such as Graph Neural Networks.

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